

A Cryptographic Infrastructure for Friend-to-Friend Choreographies

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A framework for

¹See [4, 5, 2, 1] for a formal model of local-first coordination.

A framework for

friend-to-friend [7, 8]

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coordination

and its (forthcoming) implementation

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*To appear on
JLAMP 2026*

- Motivations for re-considering distributed computing
(and not being "obscured by clouds")

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- ▶ Motivations for re-considering distributed computing
(and not being "obscured by clouds")
- ▶ A new choreographic model
 - ▶ F2F-choreographies
 - ▶ Nearbybytes

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- ▶ Motivations for re-considering distributed computing
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- ▶ A new choreographic model
 - ▶ F2F-choreographies
 - ▶ Nearbytes
- ▶ Implementation aspects

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JLAMP 2026*

Blessing and curse of clouds

Advantages of clouds

cloud computing = fast communication infrastructures + global-scale computation facilities

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- ▶ anytime-anywhere motto
- ▶ enabler of service-oriented computing/microservices (software-as-service)
- ▶ dynamic reconfiguration (services can be replaced if needed)
- ▶ reduced costs for maintenance, facilities, personnel, software, and hardware

No free lunch

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Where would you like to keep your data?
Can you control how they are used?

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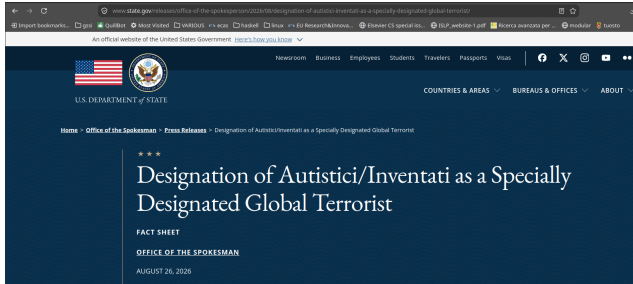
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Anytime, anywhere...really?

- ▶ *like during the AWS's outage on 25/11/2020*
- ▶ *or almost all Google services down on 14/12/2020*
- ▶ *DSL typical availability of 97%*
- ▶ *acutally, some SLA have no lower bound [What Is the Digital Divide?]*

...and limited autonomy/sovereignty



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en.wikipedia.org/wiki/Autistici/Inventati

QuillBot Most Visited VARIOUS ecas haskell linux EU Research&Innova... Elsevier CS special iss...

as Monuments: Photograph a monument, help Wikipedia and win!

more

Autistici/Inventati

Article Talk

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"A/I" redirects here. For other uses, see [AI \(disambiguation\)](#).

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Autistici/Inventati (A/I) was an Italian **hacktivist** group that provided Internet communication services for groups opposed to **fascism**, **militarism**, **racism**, **sexism**, **homophobia** and **transphobia**.^[1]

Abbreviation	A/I
Established	2010s
Dissolved	6 September 2026
Type	Non-profit
Services	Encrypted communication

...and limited autonomy/sovereignty



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QuillBot Most Visited VARIOUS ecas haskell linux EU Research&Innova... Elsevier CS special iss...

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Type	Nc
Services	En

Perhaps some re-thinking is needed?

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- ▶ kept reliance on the cloud to a minimum?
- ▶ retained control over
 - ▶ data (i.e., ensure **confidentiality**)
 - ▶ computational resources
- ▶ had more "autonomy"

What if we went friend-to-friend too?

In Li & Dabek's words [7]

F2F: reliable storage in open networks

Abstract [...] We argue that a node should choose its neighbors (the nodes with which it shares resources) based on existing social relationships instead of randomly. This approach provides incentives for nodes to cooperate and results in a more stable system which, in turn, reduces the cost of maintaining data. [...]

In Popescu, Crispo & Tanenbaum words [8]

[...]Friends Team-Up and Beat the System

Abstract [...] while existing peer-to-peer architectures with similar aims attempt to build trust relationships on top of the basic, trust-agnostic, peer-to-peer overlay, Turtle takes the opposite approach, and builds its overlay on top of pre-existent trust relationships among its users. [...]

Local-first coordination
via
Friend-to-Friend Choreography

Our choreographic model

Syntax

$$\begin{aligned} F &::= A \rightsquigarrow B : t \\ &\quad | F + F' \\ &\quad | F ; F' \\ &\quad | F \mid F' \\ &\quad | \dots \end{aligned}$$

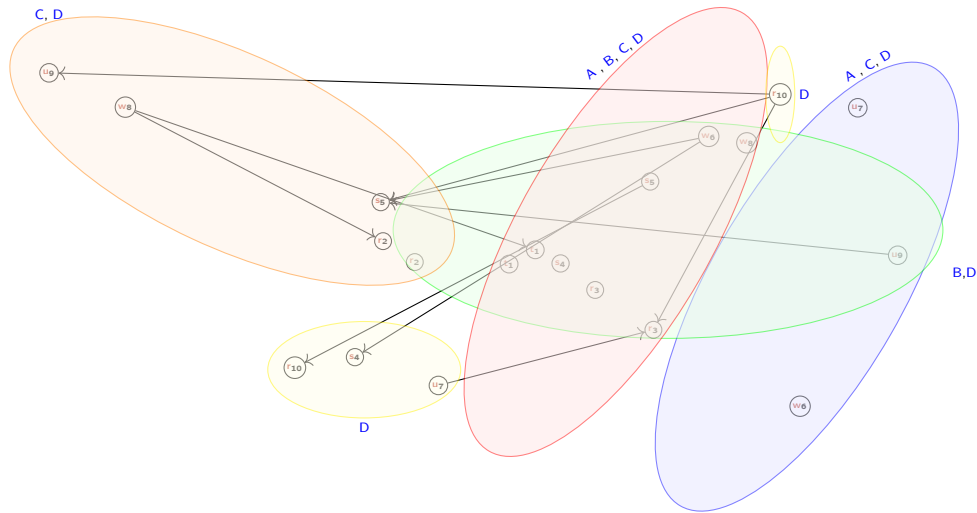
where

- ▶ B is a finite (possibly empty) set of A 's "buddies"
- ▶ t is a "type"

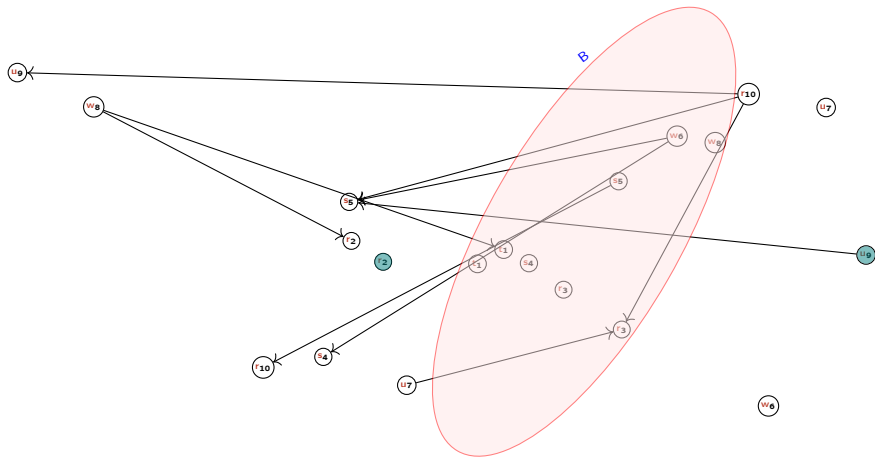
Semantics

$$\begin{aligned} \llbracket A \rightsquigarrow B : t \rrbracket &= \left\{ [A!B\langle t \rangle] \right\} \\ \llbracket F + F' \rrbracket &= \llbracket F \rrbracket \cup \llbracket F' \rrbracket \\ \llbracket F ; F' \rrbracket &= \{ r ;_{\varepsilon_r} r' \mid \dots \} \\ \llbracket F \mid F' \rrbracket &= \{ r ;_{\emptyset} r' \mid \dots \} \\ \llbracket \dots \rrbracket &= \dots \end{aligned}$$

The coordination model, informally



Replication



Having fun together

Having fun together

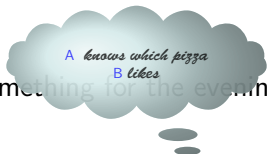
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Having fun together

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A's options are between staying home for a pizza or going out to a cinema

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- ▶ if A opts for a cinema, B decides which movie; E should be oblivious of the title, but not of the choice
- ▶ otherwise, deal is struck with whoever tells A which pizza they want
- ▶ h and c may be public, but the rest should stay confidential (e.g., X's choice of pizza is known only to X and A)

Having fun together

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since E would
complain anyway

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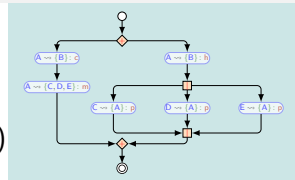
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An F2F-choreography for fun

$$F = \begin{array}{l} A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p) \end{array}$$

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$$\llbracket F \rrbracket = \left\{ \left[\begin{array}{c} A!B\langle c \rangle \\ \downarrow \\ B!\{A, C, D\}\langle m \rangle \end{array} \right], \left[\begin{array}{c} A!B\langle h \rangle \\ \swarrow \quad \downarrow \quad \searrow \\ C!A\langle p \rangle \quad D!A\langle p \rangle \quad E!A\langle p \rangle \end{array} \right] \right\}$$

Nearbytes and its cryptographic infrastructure

What is Nearbybytes?

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- ▶ set of protocols for confidential interactions combining
 - ▶ communication via "physical wiring"
 - ▶ with cryptography + "hubs" merging

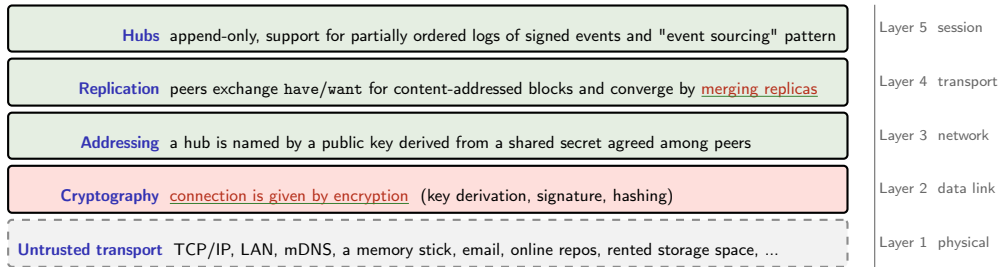
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- ▶ logical content-based addressing leveraging public-key cryptography
- ▶ data-related events form a partially ordered log

Nearbytes's stack

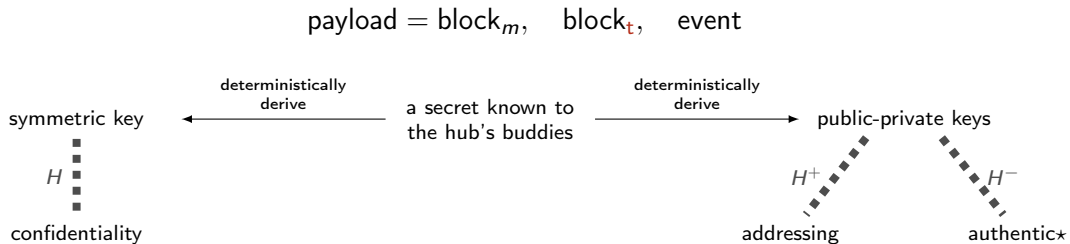


Nearbytes

Hush-hush (no key distribution)

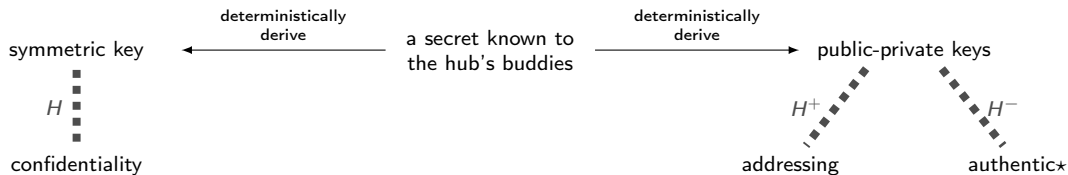
payload = block_m, block_t, event

Hush-hush (no key distribution)



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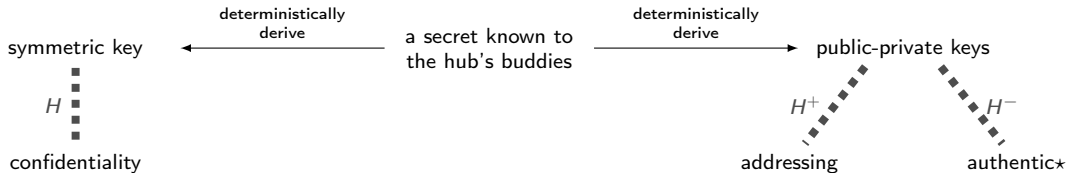
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$$\text{block}_m = \{ \text{msg} \}_H$$

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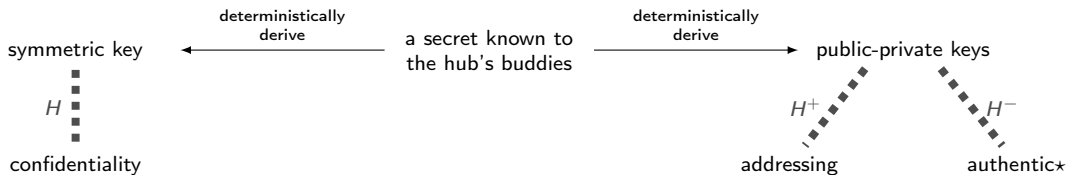
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block_m = { msg }_H
block_t = { type }_{H⁺}

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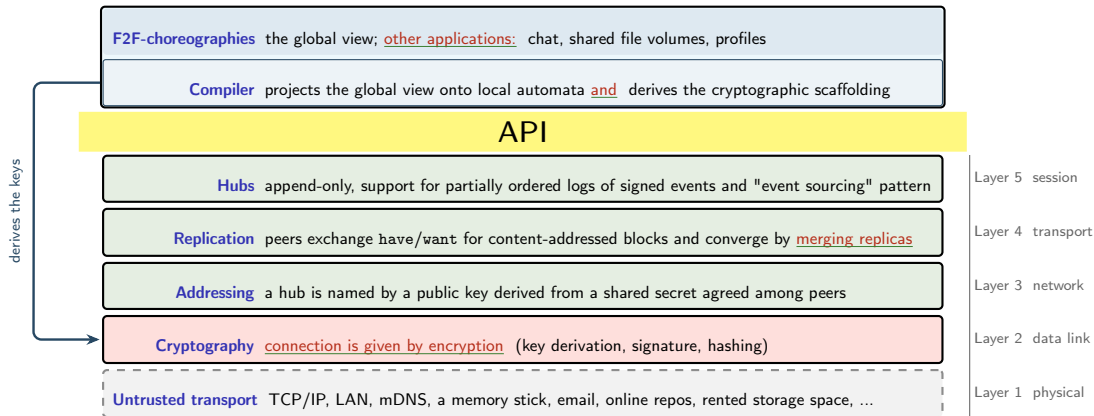
block_m = { msg }_H

block_t = { type }_{H⁺}

event = { hash(ev₁), ..., hash(ev_k), hash(block_m), hash(block_t) }_{H⁻}

F2F-choreographies & Nearbybytes

F2F-choreographies on top of Nearbybytes



Hubs' addressing

Buddies agree on a "master secret" M

Leverage buddies' public keys

$$\left\{ \begin{array}{lcl} H & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \cdot E^+ \\ H_1 & = & M \cdot A^+ \cdot B^+ \cdot C^+ \cdot D^+ \\ \dots & & \\ H_{14} & = & M \cdot A^+ \cdot B^+ \\ H_{15} & = & M \end{array} \right.$$

Projections

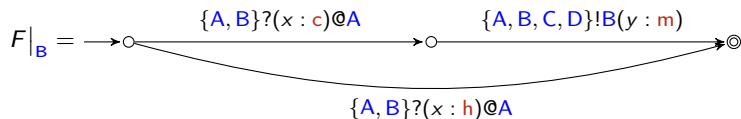
Deriving buddies' expected behaviour:

$$F = \quad A \rightsquigarrow B : c; B \rightsquigarrow \{A, C, D\} : m \\ + \quad A \rightsquigarrow B : h; (C \rightsquigarrow A : p \mid D \rightsquigarrow A : p \mid E \rightsquigarrow A : p)$$

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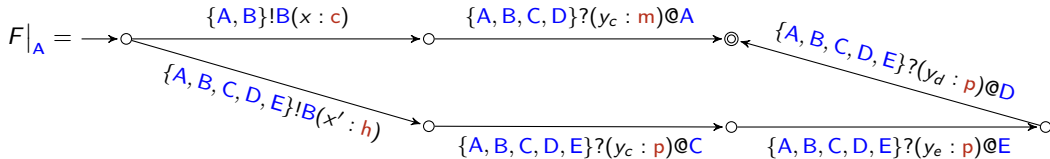
$$\begin{aligned} F = & \quad A \rightsquigarrow B : \textcolor{blue}{c}; B \rightsquigarrow \{A, \textcolor{blue}{C}, \textcolor{blue}{D}\} : \textcolor{brown}{m} \\ & + \quad A \rightsquigarrow B : \textcolor{brown}{h}; (C \rightsquigarrow A : \textcolor{brown}{p} \mid D \rightsquigarrow A : \textcolor{brown}{p} \mid E \rightsquigarrow A : \textcolor{brown}{p}) \end{aligned}$$



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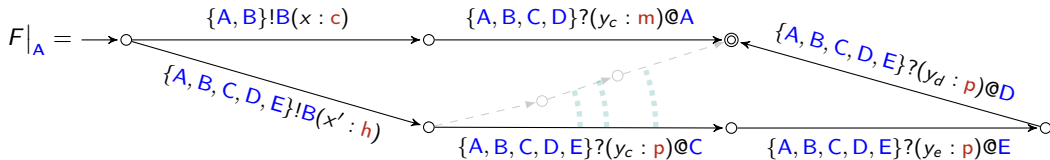
$$\left[A! \{A, B, C, D, E\} \langle h \rangle \right]$$

$$e = \{A \text{ 's address}\}_H, \{h\}_{H+}, \{\text{hash}(\{A \text{ 's address}\}_H), \text{hash}(\{h\}_{H+})\}_{H-}$$

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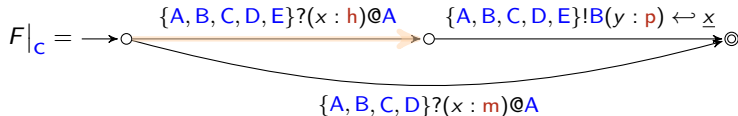
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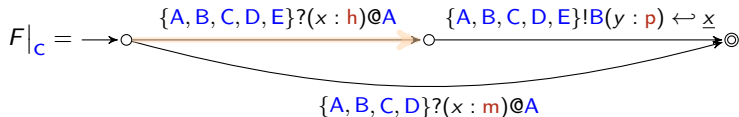
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$$\left[\begin{array}{c} A! \{A, B, C, D, E\} \langle \textcolor{blue}{h} \rangle \\ \downarrow \\ C! \{A, B, C, D, E\} \langle \textcolor{blue}{p} \rangle \end{array} \right]$$

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$$\{\text{margherita}\}_H, \{\textcolor{blue}{p}\}_{H^+}, \{\text{hash}(e), \text{hash}(\{\text{margherita}\}_H), \text{hash}(\{\textcolor{blue}{p}\}_{H^+})\}_{H^-}$$

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- ▶ Compile the automata to actual multithreaded code (e.g., Typescript or GoLang) and use `Nearbybytes` as a library

Wrapping-up

Future work

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- ▶ Consider specific notions of eventual consistency
- ▶ Extend *Nearbytes* with broadcast encryption schemes?

Thank you!

On "Obscured by clouds" I like

Bridges burning gladly
Merging with the shadows,
Flickering between the lines.
Stolen moments floating softly on the air,
Borne on wings of fire
and climbing higher.

Ancient bonds are breaking,
Moving on and changing sides.
Dreaming of a new day,
Cast aside the other way.

Magic visions stirring,
Kindled by and burning flames rise in her eyes.

The doorway stands ajar,
The walls that once were high.
Beyond the gilded cage,
Beyond the reach of ties.
The moment is at hand.

She breaks the golden band.

(Pink Floyd)

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